HAVING vs WHERE

Short Documentation: HAVING vs WHERE (TechSphere Solutions Database)

This SQL script demonstrates the difference between **WHERE and HAVING clauses** using the techsphere_solutions database. It focuses on how SQL filters data before and after aggregation.

1. Database Selection

```
USE techsphere_solutions;
```

What it does:

- Selects the active database
- Ensures all queries run on the correct schema

2. Introduction Concept (WHERE vs HAVING)

WHERE

- Filters **individual rows before grouping**
- Works on raw table data

HAVING

- Filters **grouped/aggregated results after grouping**
- Works with GROUP BY

3. GROUP BY with Aggregation

```
SELECT role_title, AVG(base_salary)
FROM job_roles
GROUP BY role_title;
```

What it does:

- Groups job roles
- Calculates average salary per role

4. Incorrect Use of WHERE with Aggregate Function

```
SELECT role_title, AVG(base_salary)
FROM job_roles
WHERE AVG(base_salary) > 100000
GROUP BY role_title;
```

What it does (and why it fails):

- Attempts to filter using AVG() inside WHERE
- Fails because WHERE runs **before grouping**
- Aggregate functions are not allowed in WHERE

5. Correct Use of HAVING

```
SELECT role_title, AVG(base_salary)
FROM job_roles
GROUP BY role_title
HAVING AVG(base_salary) > 100000;
```

What it does:

- Groups roles
- Calculates average salary
- Filters only groups where average salary is above 100,000

6. HAVING with Alias

```
HAVING avg_salary > 100000;
```

- Uses alias (avg_salary) for cleaner filtering
- Same result as previous query

7. Using WHERE + HAVING Together

```
WHERE base_salary > 60000
GROUP BY role_title
HAVING avg_salary > 100000;
```

What it does:

- WHERE filters raw salary records first
- GROUP BY organizes data by role
- HAVING filters grouped results after aggregation

8. Practical Business Example

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING total_employees > 1;
```

What it does:

- Counts employees per department
- Returns only departments with more than 1 employee
- Useful for workforce analysis

9. Key Difference Summary

Clause	Purpose
WHERE	Filters individual rows before grouping
HAVING	Filters aggregated/grouped results

10. Core Learning Outcome

After this script, you understand:

- How SQL processes queries step by step
- Why aggregate functions cannot be used in WHERE
- How HAVING enables filtering of grouped data
- How to combine WHERE, GROUP BY, and HAVING effectively

11. Summary Insight

WHERE works on raw data, while HAVING works on summarized (grouped) data. Together, they allow precise control over both filtering and analysis in SQL.